

## CY 405: Introduction to Structural Chemistry

Course Code: CY 405

Course Name: Introduction to Structural Chemistry

L-T-P-C: 3-0-0-3

Prerequisites: None

Intended for: BS

Approval: 64th BoA

### Preamble

*Course Overview:* Structural Chemistry is the important for understanding how the microscopic arrangement of atoms commands the macroscopic properties of materials. The course begins at the atomic level, exploring the fundamental forces and interactions that bind atoms together before scaling up to the periodicity of crystalline solids.

*Objectives and Scope:* Students will understand the fundamental forces and interactions governing organic molecules, coordination complexes, metals, non-metals and ionic solids. By examining crystal lattices, crystal structures, symmetry, packing motifs, and phase transitions, students will learn to predict material behavior. The course bridges abstract theories with practical application, focusing structural principles, polymorphism and intermolecular interaction with their applications pharmaceutical industry.

### Course Modules

- **Module 1: Foundation of Structural Chemistry:** Interatomic bonds, energies of chemical interaction of atoms, concept of bond angles and bond distances in (organic molecules, organometallic compounds, clusters and coordination compounds), intermolecular interactions (crystallographic van der Waals Radii, equilibrium radii of atoms, anisotropic van der Waals radii and donor acceptor interactions) (lectures 6)
- **Module 2: Crystal Lattice, crystal structures and crystal properties:** Introduction to crystallography, translational periodicity, crystal lattices and crystal structure, characteristics of crystal structure, structural motifs, crystal defects, Introduction to crystallographic space groups, symmetry in crystal structure, applications of crystal structure, Introduction to crystal packings and structures using software tools, Cambridge crystallographic data center (CCDC) (lectures 6)
- **Module 3: Structure of metals, non-metals and ionic compounds:** Packing and structure of pure metals, structure of solid solutions and metallic compounds, structure of surface of metals, structural chemistry of non-metallic compounds, structural chemistry and importance of hydrogen bonding, structural chemistry of H<sub>2</sub>O, ionic crystal structures, linkage rules in coordination polyhedral, Pauling rule of ionic crystal structure, structural chemistry of silicates. (lectures 14)
- **Module 4: Phase Transitions and Polymorphism:** Polymorphism, energies of phase transition, enthalpies of phase transformations, polymorphic transformations, effect of phase transition on crystal properties. (lectures 6)
- **Module 5: Applications and case studies in pharmaceuticals** (lectures 5)

### References

1. *Crystal Engineering:* G. R. Desiraju by World Scientific Publishers.

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2. *Perspective in Structural Chemistry*, J. K. Burdett, Chem. Rev. 1988, 88, 3-30.

#### **Text Books**

1. *Fundamental of structural Chemistry* by Gong-Du-Zhou; publisher world scientific.
2. *Introduction to structural chemistry* by Stepan S. Batsanov, Andrei S Batsanov; publisher Springer.